9.2.1 Singular matrices and the eigenvalue problem

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Week 09 due Mar 19, 2025 07:24 IST

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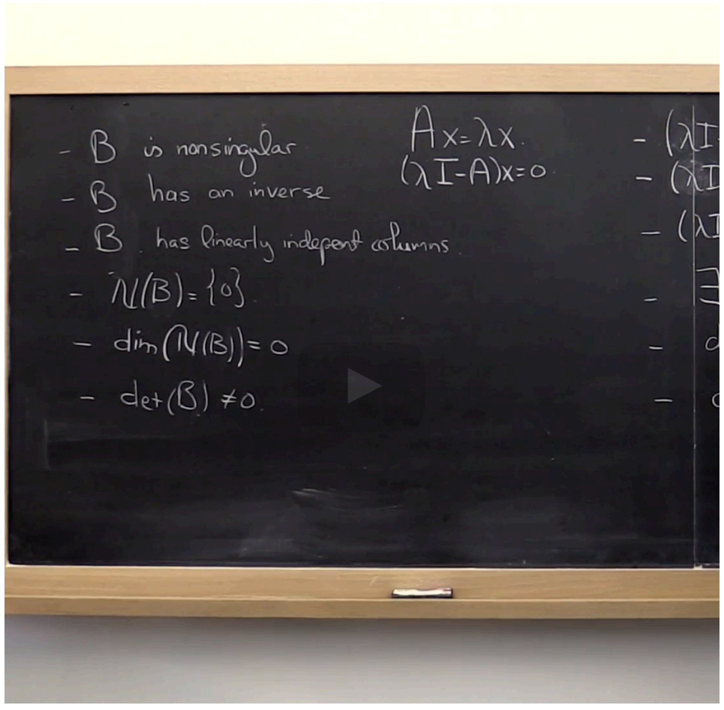

contains the zero vector, it's dimension of the null space

is equal to zero and the determinant B is not equal to zero. And as a matter of fact,

there was a whole list of these kinds of equivalent statements. And it turns out

that as we talk about eigenvalue problems, eigenvalues and eigenvectors,

what we'll do is we'll strategically pick the equivalent statement that is most

appropriate for a

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Reading assignment

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Read Unit 9.2.1 of the notes. [LINK]

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matrix A. Okay. And that's known as the spectrum of

matrix A. The spectral radius of matrix A is defined as the largest absolute value

of any of the eigenvalues of matrix A. Okay?

So why do we call this the spectral radius. Well, you can think of all of the

eigenvalues, they're scalars, and therefore they are complex valued numbers, well

real or complex values. And if you put them all in the complex plane, and you

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diagonal elements, well the

absolute values of the off diagonal elements in the row indexed with 0. If you

now go to the point alpha_00 in the complex plane, and you draw a circle

around that with radius rho_0 defined as such. And you do that for all of the

diagonal elements and the corresponding radii -- so here we have another disk of

radius rho_1 and another disk of radius rho_2 and another disk of radius rho_3.

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matrix A of zero. Okay? Then we can start varying our omega. And an insight is that

where the eigenvalues of matrix A are varies continuously with this parameter

omega. And that's you know that's another argument. But as far as the Gershgorin

disk theorem is concerned, what this means is that when omega is very small

these disks are very small. And because of this continuity argument what we

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Homework 9.2.1.1

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times m}$.

TRUE/FALSE: $0 \in \Lambda(A)$ if and only A is singular.

TRUE ✔ Answer: TRUE

Solution

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 9.2.1.2

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times m}$ be Hermitian.

ALWAYS/SOMETIMES/NEVER: All eigenvalues of A are real-valued.

ALWAYS ✔ Answer: ALWAYS

Solution

For a complete solution, see the notes .

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Homework 9.2.1.3

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be Hermitian positive definite (HPD).

ALWAYS/SOMETIMES/NEVER: All eigenvalues of A are positive.

ALWAYS ✔ Answer: ALWAYS

Solution

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 9.2.1.4

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be Hermitian, (λ, x) and (μ, y) be eigenpairs associated with A, and $\lambda \neq \mu$.

ALWAYS/SOMETIMES/NEVER: $x^H y = 0$

ALWAYS ✔ Answer: ALWAYS

Solution

For a complete solution, see the notes .

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Homework 9.2.1.5

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$, (λ, x) and (μ, y) be eigenpairs, and $\lambda \neq \mu$. Prove that x and \(y) are linearly independent.

☑ done

✔

Solution

For a complete solution, see the notes .

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Homework 9.2.1.6

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times m}$, $k \le m$, and (λ_i, x_i) for $1 \le i < k$ be eigenpairs of this matrix. Prove that if $\lambda_i \neq \lambda_j$ when $i \neq j$ then the eigenvectors x_i are linearly independent. In other words, given a set of distinct eigenvalues, a set of vectors created by taking one eigenvector per eigenvalue is linearly independent.

☑ done

✔

Solution

For a complete solution, see the notes .

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Homework 9.2.1.7

1/1 point (graded)

Consider the matrices

$$A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$$

and

$$\left(\begin{array}{cccc|cccc|cc} 4 & -1 & & & -1 & & & & & \\ -1 & 4 & -1 & & & -1 & & & & \\ & -1 & 4 & -1 & & & -1 & & & \\ & & -1 & 4 & & & & -1 & & \\ \hline -1 & & & & 4 & -1 & & & -1 & \\ & -1 & & & -1 & 4 & -1 & & & \ddots \\ & & -1 & & & -1 & 4 & -1 & & \\ & & & -1 & & & -1 & 4 & & \\ \hline & & & & -1 & & & & 4 & \ddots \\ & & & & & & \ddots & & \ddots & \ddots \end{array}\right)$$

ALWAYS/SOMETIMES/NEVER: All eigenvalues of these matrices are nonnegative.

ALWAYS ✔ Answer: ALWAYS

ALWAYS/SOMETIMES/NEVER: All eigenvalues of the first matrix are positive.